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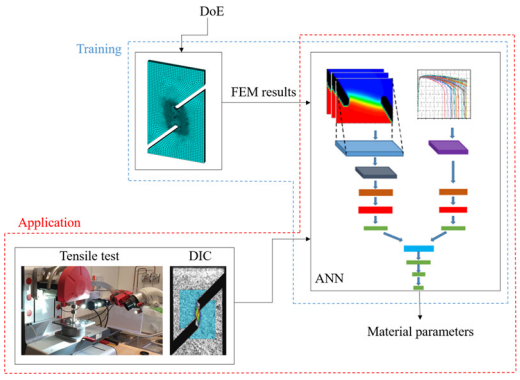
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HIGHLIGHTS

- Machine learning supported material characterisation using one single specimen.
- Training data includes displacement field evolution extracted from the finite element model.
- Local stress state evolution is implicitly contained in the training of the artificial neural network.
- Facilitated calibration of stress state dependent and regularized damage model.

GRAPHICAL ABSTRACT



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ABSTRACT

While several different specimen geometries are typically required to calibrate a ductile fracture locus model, this article presents for the first time a calibration methodology that uses one single specimen geometry. This is accomplished by a computational framework that combines finite element modelling (FEM) and artificial neural network (ANN). The combinations of the model parameters are used to generate the training database. The local displacement fields and global force-displacement histories are extracted throughout the complete numerical experiment and passed to the ANN. Therefore, the influence of the local stress state on the evolution of the local deformation is implicitly taken into account. The trained ANN is verified by evaluating its predictability of material parameters of FE simulations unseen in the training stage. The experimental data obtained from the shear tensile test using Digital Image Correlation is introduced to the trained ANN to identify the parameter set that predicts the real mechanical response of the shear specimen. Three different ANN architectures with distinguished input representations are studied. It turns out that all of them can acceptably describe the experimental behaviour of not only the calibration specimen but also the specimens not used for training the model.

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1. Introduction

Ductile damage and fracture modelling is essential for a broad spectrum of metal alloys applications, ranging from large lightweight structures over car bodies to tiny electronic components. Although sophisticated stress state dependent damage models have been proposed in the literature, the complexity of the calibration procedure often hinders their application in the aerospace, automotive and electronics communities.

For its simplicity and ease of implementation, uncoupled fracture initiation models are convenient for industrial purposes and widely

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used in applied mechanics, among others [10,16–18,20–22,26,29,32]. Their heart is a fracture locus that quantifies the dependency of ductile fracture on the stress and strain states [10]. [8] took a step further and embedded the fracture locus concept into a coupled damage mechanics framework. Either used in uncoupled or coupled mode, ductile fracture models require knowledge of the strain to fracture as a function of the stress state for proportional loading for their straightforward calibration. An ideal case of a constant strain-history path is possible to achieve just in special cases [32,33]. As imposing constant proportional loadings up to failure over the complete stress state domain is extremely difficult, if not impossible, determination of the actual fracture locus from experimental measurements becomes questionable.

In addition to the experimental challenges mentioned above, there are uncertainties related to the finite element (FE) simulations which are indispensable for extracting the evolution of mechanical fields in a material point, i.e. the progress of the stress triaxiality and Lode angle parameter upon straining. Each experiment has to be studied by FE analysis to identify the stress and strain fields. [18] demonstrated that spatial and time discretization errors might greatly affect the evolution of both equivalent plastic strain and stress triaxiality. Furthermore, the location of the onset of fracture in the computational model is frequently assumed to coincide with the location of the highest equivalent plastic strain. However, because of the stress state effects, the most strained element does not necessarily have to reach the critical damage value first which defines the instant of the onset of fracture in a FE model. Furthermore, the analysis of the mechanical fields discussed beforehand often relies on damage-free plasticity, although there is ample evidence that coupled damage plasticity models provide more realistic results in structural computations.

In this work, a machine learning supported calibration of a ductile fracture locus model is investigated in an attempt to overcome the above challenges in experimental and computational mechanics. Several studies already testify the great potential of artificial neural network (ANN) systems for material parameter identification. For instance, [2] followed the ANN-based identification strategy to calibrate the parameters of the Gurson-Tvergaard-Needleman (GTN) model [19,36] by using the X and Y displacements of the specimen's middle point. [3,4] employed the load-displacement curve of a small punch test to train the ANN and find the damage and fracture properties modelled by the GTN model as well. [5,28] used an ANN to find the parameters of the GTN model employed in sheet metal blanking simulations. More recently, [35] also used the information from the load-displacement curve for ANN training to determine the damage parameters of the shear modified GTN model [30].

We present a computational framework for the determination of material constants by a machine learning method that integrates finite element (FE) simulations with ANN, similar in purpose but significantly different from the previous works in the following aspects:

- training data selection: the data passed from FE to ANN model consists of (i) the images of a complete local displacement field and (ii) the values of the global force-displacement curve. Both (i) and (ii) data are extracted throughout the complete numerical experiment;
- material model employed: a machine learning supported calibration of a ductile fracture locus model will be attempted for the first time in the following sections.

The main advantage of the proposed calibration strategy stems particularly from the above specific training data selection. While using only one single calibration specimen, the presented approach allows to make use of all information accessible in the experiment, i.e. the local deformation influenced by the stress state and the damage state. In particular, assisting the calibration of a specific fracture locus model by ANN allows to circumvent extracting and averaging the evolution of stress state variables in a material point from numerical models of various calibration specimens. Therefore, no assumptions regarding the choice of an element for extracting the loading paths need to be adopted any longer.

The framework features (1) training of the ANN merely from numerical experiments, (2) examination of the ANN model's quality by assessing the predictability of material parameters used in FE simulations unseen in the training stage, (3) feeding the experimental data to the trained ANN to find the set of parameters that models the behaviour of real experiment and (4) validation on different sample geometries not used for the training.

The goal of the present study is to find a robust set of parameters that accurately models the mechanical behaviour and the instant of fracture of the analysed ductile material. We limit our attention to the parameters of the coupled damage model based on the fracture locus introduced in the preceding work [8]. Instead of performing a series of ductile fracture experiments on various sample geometries to characterize the macroscopic mechanical response of the investigated alloy, a single sample geometry is analysed numerically by FE simulations. A shear tension specimen is modelled to study the sensitivity of the numerical response to parameter variations selected by a three-level full factorial experimental design. The data extracted from the FE models that are passed to the ANN will be referred to as a *training database*. The ANNs used here belong to the class of convolutional neural networks (CNN), which are chosen because of their pattern recognition capability. In particular, CNNs are able to exploit translation and scale invariance, making them generally more data-efficient for most pattern recognition tasks. The ANN key features include one dimensional (1D) and three dimensional (3D) convolution layers with subsequent max pooling, and, in addition, two regularization approaches applied together to mitigate the overfitting of the parameters to be estimated.

2. Constitutive model: Governing equations

The coupled ductile fracture locus model employed in the sequel is briefly summarized below to the reader's convenience. Details on the model formulation can be found in [8]. Since the fundamental mechanisms of ductile fracture are rather well understood, cf., e.g., the review articles [13,31], the constitutive equations are preserved to keep the physical foundation of the model. The function equations are thus predetermined, and the ANN is used for determining the constants of this given function.

2.1. Coupled ductile fracture locus model

The unified approach introduced in [8,9] comprises the theories for ductile fracture locus [27], damage mechanics [25] and regularization [6]. The model proposes modelling strain softening linked to damage evolution dependent on the current stress state along the strain path. The equivalent stress required for yielding of the damaged material matrix is described by

$$\sigma_{eq} = \left[1 - \int_0^{\varepsilon_f} \frac{d\varepsilon_p}{\varepsilon_f \eta, \bar{\theta}}\right] \left[\sigma\varepsilon_p + L\,\varepsilon_p - e_p\right], \tag{1}$$

where a three-dimensional fracture locus $\varepsilon_f(\eta, \bar{\theta})$ defines a function that returns the fracture strain ε_f at a general multiaxial stress state given by the stress triaxiality η and the Lode angle parameter $\bar{\theta}$. A linear damage accumulation rule, $\frac{d\varepsilon_p}{\varepsilon_f(\eta,\bar{\theta})}$, where $d\varepsilon_p$ denotes the differential increment of the equivalent plastic strain ε_p, quantifies the local ductile damage D assumed to accumulate at each point in the continuum prior to failure. A softening factor is defined as a linear function of the damage variable, $[1 - \int_0^{\varepsilon_f} dD]$, to capture the stress state effects on the plastic flow, $\sigma(\varepsilon_p)$. An additional regularization term, $L\,(\varepsilon_p - e_p)$, modifies a conventional hardening function $\sigma(\varepsilon_p)$ in an additive manner. Note that the regularization is applied to the current yield strength and not to the initial undamaged one. The variable e_p represents a weighted average of

Table 1
Material parameters to be identified by the ANN.

Ductile fracture locus parameters	c_1 [−] 2.082[a]	c_2 [−] ?	c_3 [−] ?
Regularization parameters	l [mm] 0.195[b]	L [MPa] ?	

[a] c_1 is fixed and omitted from the ANN; cf. the main text for a detailed discussion.
[b] The length scale l is determined from experiments as the width of the shear band observed by Digital Image Correlation [8].

the equivalent plastic strain ε_p over a specified strain distribution function f that follows a Gaussian distribution. The implemented regularization distributes the strains of the FE model over a width defined by a length scale parameter l contained in the strain distribution function f. As the influence of the softening and regularization terms interfere, the parameter L should be calibrated to produce an adequate amount of regularization.

The linear damage accumulation rule is a simple form of mapping all possible fracture surfaces obtained from different strain paths onto a single fracture locus given by [27],

$$\varepsilon_f(\bar{\theta},\eta) = \frac{c_3}{\left(\frac{2}{\sqrt{\bar{\theta}^2+3}}\right)^{c_1}\left(\frac{\langle 1+3\eta\rangle}{2}\right)^{c_2}}, \langle x\rangle = \begin{cases} x & when\ x \geq 0 \\ 0 & when\ x < 0 \end{cases} \quad (2)$$

where the material constants $\{c_1, c_2, c_3\}$ should be determined. c_1 and c_2 model the stress state dependency of the underlying damage process, whereas c_3 represents the magnitude of the fracture locus without influencing its final shape. The fracture strain, ε_f, is herein defined as the maximum local equivalent plastic strain just before structural failure.

2.2. Unknowns for the ANN model

The three parameters $\{c_1, c_2, c_3\}$ of the fracture locus defined by (Eq. 2) are to be found from a shear tension experiment. As c_1 reflects the influence of the Lode angle parameter, $\bar{\theta}$, whose variation in the shear tension experiment (see subchapter 3.1.) is expected to be minimal, it cannot be determined properly from this experiment. Accordingly, c_1 will be fixed in the following to $c_1 = 2.082$, the value found in [8]. Thereby we restrict ourselves to only two unknowns $\{c_2, c_3\}$. Given that c_2 modulates the dependency of the fracture strain on stress triaxiality, whose effect is predominant, c_2 is subject to ANN identification together with c_3, which controls the overall magnitude of the fracture strain. The regularization parameter L controlling the effect of added (artificial) hardening is to be identified by ANN as well because it has a strong impact on the strain localization. For the latter purpose, the complete strain distribution is accessible to the ANN in the form of the full surface displacement field evolution that is fed into the model.

Accordingly, we proceed with 3 unknown parameters to be identified by ANN, i.e. $\{c_2, c_3, L\}$. The fracture locus and regularization parameters of the material model employed in the sequel are summarized in Table 1. The remaining model parameters, i.e. elasticity as well as plasticity parameters defining the hardening function $\sigma(\varepsilon_p)$ (see Eq. 1), are taken from [8].

3. Computational framework for material parameter identification

3.1. Finite element model of the shear tension experiment

The finite element model of the shear specimen comprises one-half of the specimen geometry, as appropriate symmetry conditions in xy-plane are applied (see Fig. 1). The model is discretized by 8-node solid elements (ANSYS Element Library SOLID185) with an element length of about 0.043 mm and 7 elements over the thickness. The model's total thickness is 0.28 mm, which equals one half of the thickness of the real specimen machined from sheet material. The displacement history measured experimentally is applied to the upper boundary of the model and the simulations are carried out using the FE code ANSYS Mechanical APDL [7]. The choice of the parameter sets of the material model employed is further discussed in the following subchapter.

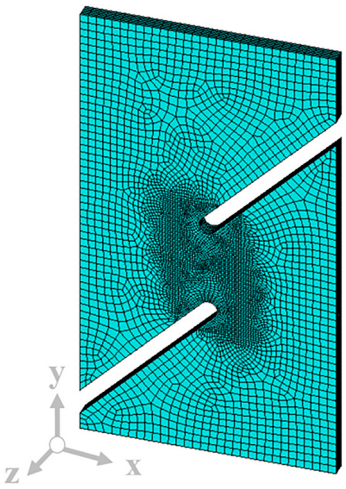


Fig. 1. Finite element model of the shear tension specimen.

In particular, this specimen was selected among the others (notched tension and dog-bone) because a closer look at the evolution of mechanical fields in the shear specimen reveals that at least one stress state parameter, i.e. the Lode angle, remains constant for most of the strain path [8]. The stress triaxiality increases from low (0.00) over moderate (0.25) to high values (0.60); a detailed analysis of the discussed loading path can be found in [8]. Being able to omit the influence of the Lode angle (and skip the determination of c_1), and having a large range of stress triaxiality and total fracture strain values for determining c_2 and c_3, is expected to facilitate the learning process of the ANN.

3.2. Training database

The sensitivity of the numerical response to material model parameter variations is studied following a three-level full factorial experimental design. The relationships between parameters $\{c_2, c_3, L\}$ are explored through finite element analysis of 27 models in total, representing 27 different positions in parameter space, arranged on a rectilinear grid, with 12 middle-edge positions (nodes), 12 corner nodes and 3 center nodes, as shown schematically in Fig. 2. Although there are somewhat more efficient designs like e.g. [15] requiring fewer simulation runs than the full factorial one, the latter was selected particularly because it covers just about every possible parameter combination. The ANN model is thereby supplied with high quality data. Yet, a reasonable parameter range has to be chosen to ensure that the actual parameter values for modelling the experimental behaviour lie within the limits of the training data. For every parameter, three levels are used, (i) the low value, (ii) the middle value and (iii) the high parameter value (Table 2). The selected range has to ensure that the domain of design space where the actual experiment falls is captured. Here, the choice of the design space was guided by the preceding work [8]. Starting from the values found in the referred work, the design variables are varied systematically and iteratively to obtain the training database well-distributed around the actual experiment, see Fig. 3.

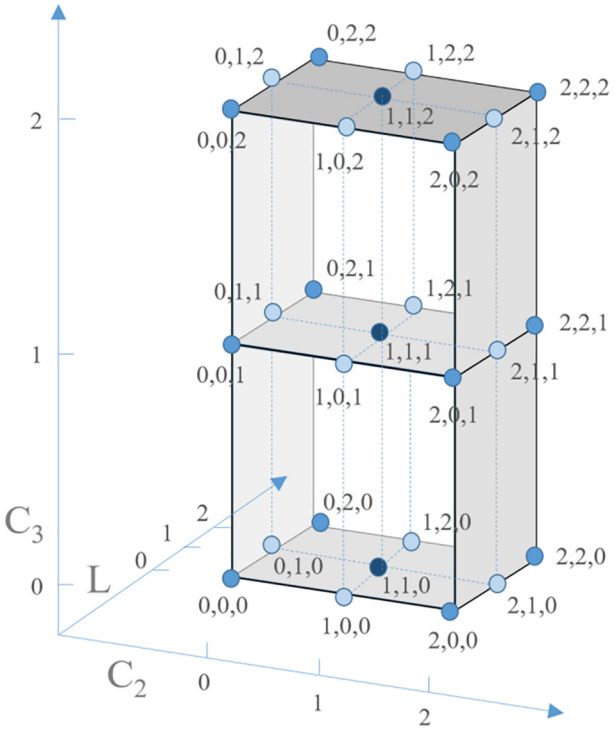


Fig. 2. Design of numerical experiments following the three-level full factorial: 3-variable system with 3 levels (0 = low value, 1 = middle value, 2 = high value) arranged on the middle edge nodes ( ), corner edge nodes ( ) and centre nodes ( ).

Table 2
ANN training database developed following a three-level full factorial design.

Numerical experiment	Ductile damage model parameters		
	Fracture locus		Regularization
	c_2 [−]	c_3 [−]	L [MPa]
Model #1	0.3	2.2	160.0
Model #2	0.5	2.2	160.0
Model #3	0.8	2.2	160.0
Model #4	0.3	2.2	180.0
Model #5	0.5	2.2	180.0
Model #6	0.8	2.2	180.0
Model #7	0.3	2.2	210.0
Model #8	0.5	2.2	210.0
Model #9	0.8	2.2	210.0
Model #10	0.3	2.5	160.0
Model #11	0.5	2.5	160.0
Model #12	0.8	2.5	160.0
Model #13	0.3	2.5	180.0
Model #14	0.5	2.5	180.0
Model #15	0.8	2.5	180.0
Model #16	0.3	2.5	210.0
Model #17	0.5	2.5	210.0
Model #18	0.8	2.5	210.0
Model #19	0.3	2.8	160.0
Model #20	0.5	2.8	160.0
Model #21	0.8	2.8	160.0
Model #22	0.3	2.8	180.0
Model #23	0.5	2.8	180.0
Model #24	0.8	2.8	180.0
Model #25	0.3	2.8	210.0
Model #26	0.5	2.8	210.0
Model #27	0.8	2.8	210.0

Postprocessing of the finite element simulations allows extracting the numerical results in terms of (a) images of the surface displacement fields in the X- and Y-direction and (b) force values for the applied displacement. The extraction of both local (a) and global (b) data is done throughout the complete numerical experiment in equally spaced time increments, Δt. In this way, the information about the stress state evolution is taken into account. Local displacement field images are extracted from the nodes positioned in the zone of interest where the plastic strain develops, see Fig. 3a. Every single force value extracted at a specific time t (Fig. 3b) is assigned to the corresponding displacement image (Fig. 3a). The database constructed thereby is passed from the FE to the ANN model, whose architecture will be discussed in the next subchapter.

The quality of the trained ANN model will be verified using the numerical experiments not contained in the training database, referred to as FE test cases. This verification is performed on FE test cases whose parameters are chosen to lie inside the design space shown in Fig. 2. The original design space is divided into 8 cubes as indicated by dashed blue lines in Fig. 2. The first set of parameter combinations is selected such that it covers the middle nodes of each of the 8 cubes (Table 3a). The second set of parameters is selected following the Latin hypercube sampling, i.e. the points inside each of the 8 cubes are chosen randomly (Table 3b).

3.3. Artificial neural network (ANN) models

3.3.1. Model architectures and the building blocks

Three ANN structures are constructed and investigated based on the way the input information is processed to estimate the material parameters, see Fig. 4. In the architecture A the local displacement field images are used to estimate the material parameters $\{c_2, c_3\}$ while the parameter $\{L\}$ is estimated by using only the information inherent in the global force-displacement profile. For the architecture B, the opposite of architecture A is imposed, i.e. the parameters $\{c_2, c_3\}$ are learned from the force-displacement profile while {L} is estimated from the displacement field images. For the architecture C all information, i.e. the features of the local displacement field images and the global force-displacement curves are used together to estimate the parameters $\{c_2, c_3, L\}$.

Since the input data for the ANNs contain images, we employ convolutional neural networks (CNN) because of their performance for translation invariant pattern recognition (in our case, the pattern to be recognized and characterized is the local strain concentration pattern). CNNs are capable of extracting high-level features hidden in the input data through their convolution layers and nonlinear activation functions. The convolutional layers are particularly suitable to exploit translation invariance in the input space. Therefore, all ANN architectures contain a convolution layer as a first layer for processing the input information. Furthermore, three-dimensional (3D) convolution layers are employed because the local displacement field images contain two space dimensions and one time dimension. The 3D convolution layers are capable of extracting features from both the spatial and temporal dimensions, which are necessary for capturing the motion information encoded in multiple adjacent frames (herein, those are data extractions on the force-displacement (FD) curve trajectory). One-dimensional convolutions are also applied on the FD history, which is a sequence of data in time. Similar to 3D convolutions, 1D convolutions are capable of extracting features from sequences of fixed length and mapping their internal features. The building blocks of the model structures shown in Fig. 4 are briefly described in the following.

In all models, after a 3D convolution layer a max-pooling layer is added, which extracts the points with the largest value in the analysed local domain. Max pooling is a non-linear operation that reduces the dimension of the features extracted by a 3D convolution and also may prevent overfitting to a certain extent. On the other side, on the FD branch, after a 1D convolution no pooling layer is added because it is not necessary due to the small feature dimension. Afterwards, a flattening layer is applied to convert the feature data of

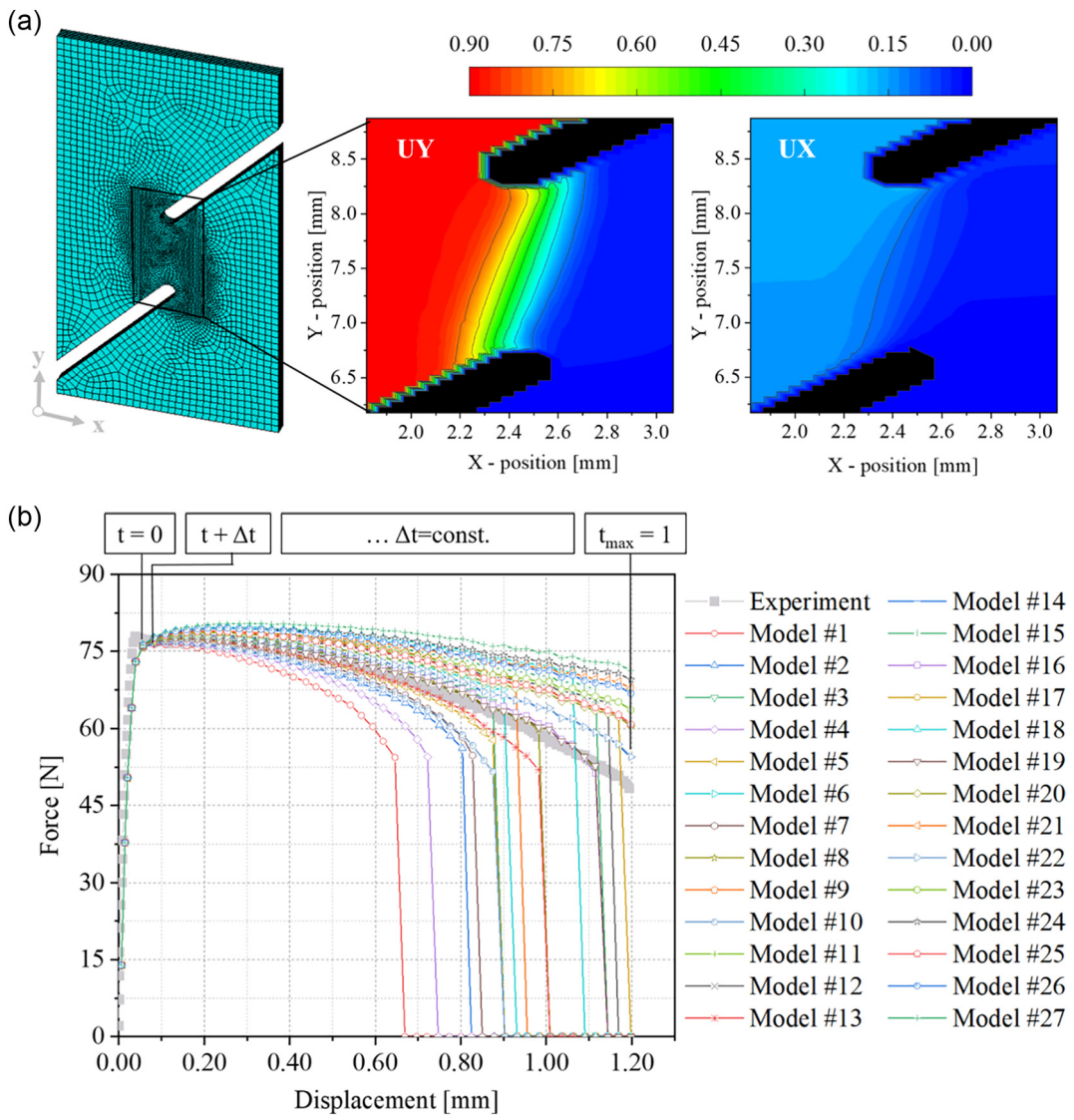


Fig. 3. Passing the training database from FE to ANN throughout the complete numerical experiment: (a) images of the local displacement field [mm] in X and Y direction (UX and UY) for every time step t, (b) global force-displacement curve values for every time step t.

the convolution layers into a single long feature vector. The structure and hierarchy of the selected layers in the CNN network follows a routine practice. For an in-depth discussion of the CNN networks, the reader is referred to [24,34].

The training data are comprised of 27 FE models, see Table 2. Considering the size of the ANN networks, the optimization problem is very much prone to overfitting, which will result in deteriorated validation or test qualities. Therefore, two regularization approaches are considered and applied together to mitigate overfitting. The first one is an additive zero-centered Gaussian noise layer, with a standard deviation of 0.05, which is positioned after the convolutional layers and can be regarded as random data augmentation. The effect of the Gaussian

Table 3a
FE test cases covering the middle points of each of the 8 cubes.

Numerical experiment	Ductile damage model parameters		
	Fracture locus		Regularization
	c_2 [−]	c_3 [−]	L [MPa]
FEM test #1	0.40	2.35	170.00
FEM test #2	0.65	2.35	170.00
FEM test #3	0.40	2.35	195.00
FEM test #4	0.65	2.35	195.00
FEM test #5	0.40	2.65	170.00
FEM test #6	0.65	2.65	170.00
FEM test #7	0.40	2.65	195.00
FEM test #8	0.65	2.65	195.00

Table 3b
FE test cases selected according to the Latin hypercube sampling.

Numerical experiment	Ductile damage model parameters		
	Fracture locus		Regularization
	c_2 [−]	c_3 [−]	L [MPa]
FEM test #9	0.32	2.43	161.63
FEM test #10	0.78	2.29	171.10
FEM test #11	0.50	2.48	184.81
FEM test #12	0.70	2.38	197.30
FEM test #13	0.31	2.50	170.33
FEM test #14	0.52	2.62	177.54
FEM test #15	0.43	2.52	200.76
FEM test #16	0.54	2.57	191.62

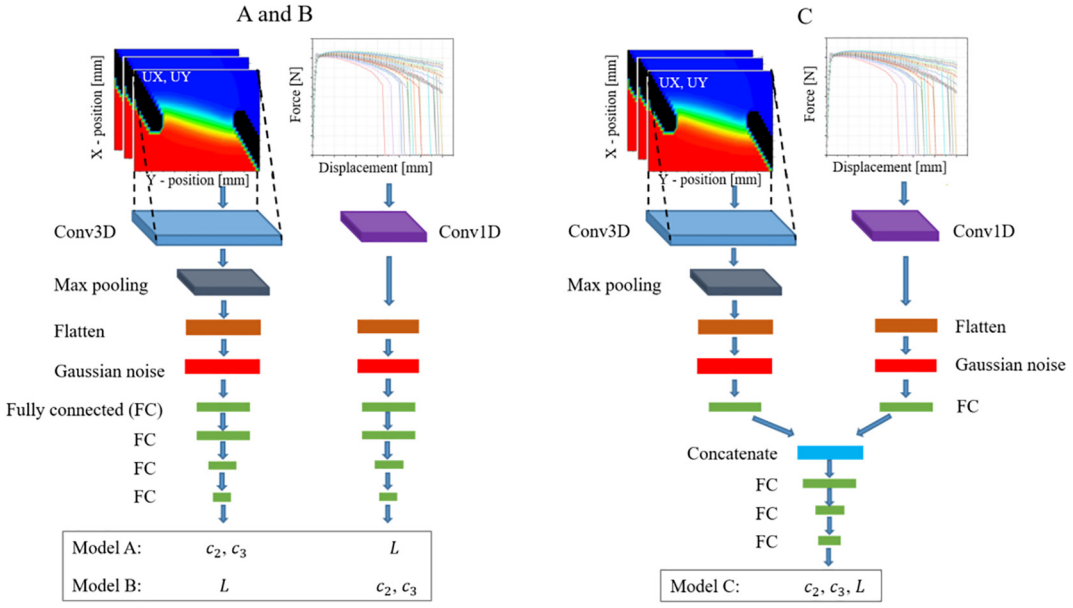


Fig. 4. Different ANN architectures studied: models A, B and C.

noise on the performance of CNNs has not been investigated systematically in the literature. However, it turned out to be very useful in achieving more generalizable trained models thereby even increasing the stability of the training stage (see the Appendix). The other regularizer is applied through penalties on all layer kernels during optimization. For the latter, we apply an L2 regularization penalty with a regularization factor of 0.01. The penalty of the L2 regularizer is computed as

$$cost = l2 \cdot \sum w^2 \tag{3}$$

where w is a vector of layer weights and $l2 = 0.01$ is the regularization factor. This penalty prevents the weights of a network to grow very large and therefore helps to minimize overfitting of the training data. After a noise layer, fully connected (FC) layers are inserted in an inverted pyramidal stack (larger layers up and smaller layers at the bottom) to emphasize the importance of the information at higher levels (see Fig. 5 for detailed information about the layer sizes). Note that in model C a concatenation layer is needed to merge two sources of extracted features to be forwarded to the next FC layers. The max-pooling, Gaussian noise, flattening and concatenation layers have no training parameters. Therefore, the model architectures A, B and C are not only similar in structure but also very close in size, i.e. models A and B have 286,816 training parameters while model C has 286,751 (see Fig. 5).

In Fig. 5 the first dimension (None) corresponds to the batch size, which for the training data is equal to 27 and for verification data is equal to 16. For the 3D and 1D CNN, filter sizes of (3,3,2) and 2 are chosen, respectively. This selection was based on the sparse random exploration of a huge hyperparameter space. The sufficiency of the kernel size 3*3 is proven in the Appendix. The spatial kernel size of 3*3 is a frequent choice as it adds more benefits like weight sharing and lowering the computational costs while taking into account information from the neighbouring pixels. The odd size of the filter (in spatial dimensions) is also beneficial because of the symmetry, i.e. all the previous layer pixels are symmetrical around the output pixel. Therefore, distortions typical for even sized kernels are avoided.

Note that these model structures do not have a large depth and width but still contain the main building blocks necessary for effective extraction of high-dimensional features from a small set of training examples. Usually, more CNN layers can potentially improve the accuracy of the ANN models. However, this depends on the choice of the architecture (deep or shallow) and depth and width of the training data set, as it has been explored in [11]. According to [11], for the training datasets of size 27, a shallow model with more FC layers is the optimal choice for better performance. Indeed, we found that adding more CNN layers does not provide extra benefits while increasing computational costs. Here, the considered one CNN layer serves as a smart feature extractor both in space and time. Finding the optimum model architecture is not trivial. However, in the context of this work, the selected architecture designs perform well enough to demonstrate the applicability of the presented approach.

3.3.2. Data preparation

Scale and distribution of the input and output data are very important for training the neural networks. Different scales may cause difficulties in training and modelling of the problem. This is because large input values may result in a model with extremely large neuron weights, thereby causing unstable learning and poor performance during generalization. Very large and different scales of output variables also often result in exploding gradients leading to a failing learning process or low model performance. Moreover, scaled training data always reduce the training time and lower the probability to fall into local minima.

In this work, the inputs (displacement field images and global FD curve) and outputs (material parameters) are normalized to the range of [0,1] with the linear transformation

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}, \tag{4}$$

where X and X' are the original and transformed data, respectively, and X_{min} and X_{max} correspond to the minimum and maximum value of X in the training data. The same minimum and maximum is used for normalization of the verification and experimental data.

3.3.3. Activation functions

In neural networks, the nonlinearity to the output of a neuron is introduced via an activation function. The five most widely used activation functions are Sigmoid, Tanh, ReLU, Softplus and Gaussian. It is not trivial a priori which activation function works best for a given data set and problem. Sigmoid and Tanh are saturated non-linear activation

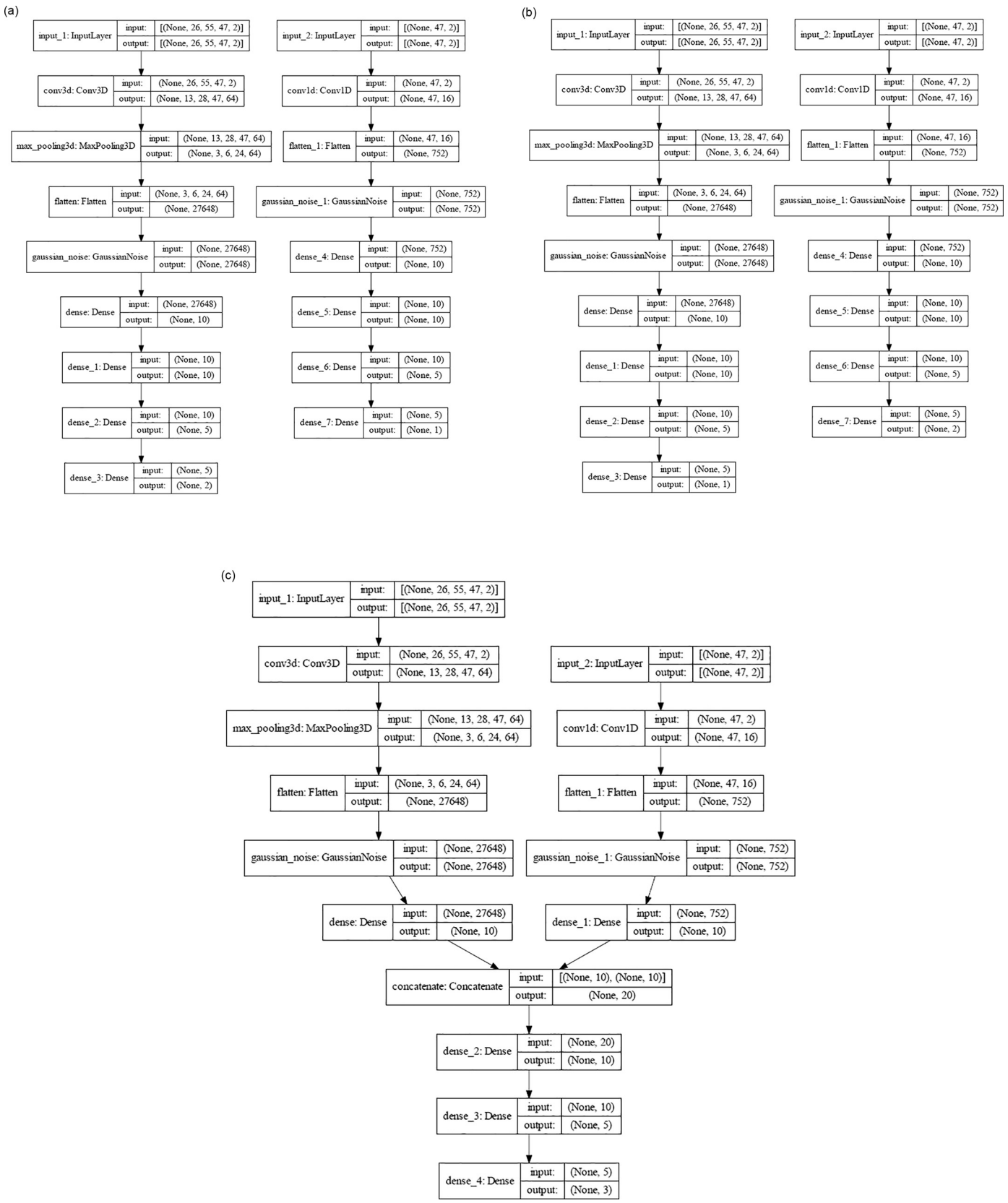


Fig. 5. Model summary of different ANN architectures: (a) model A, (b) model B, (c) model C.

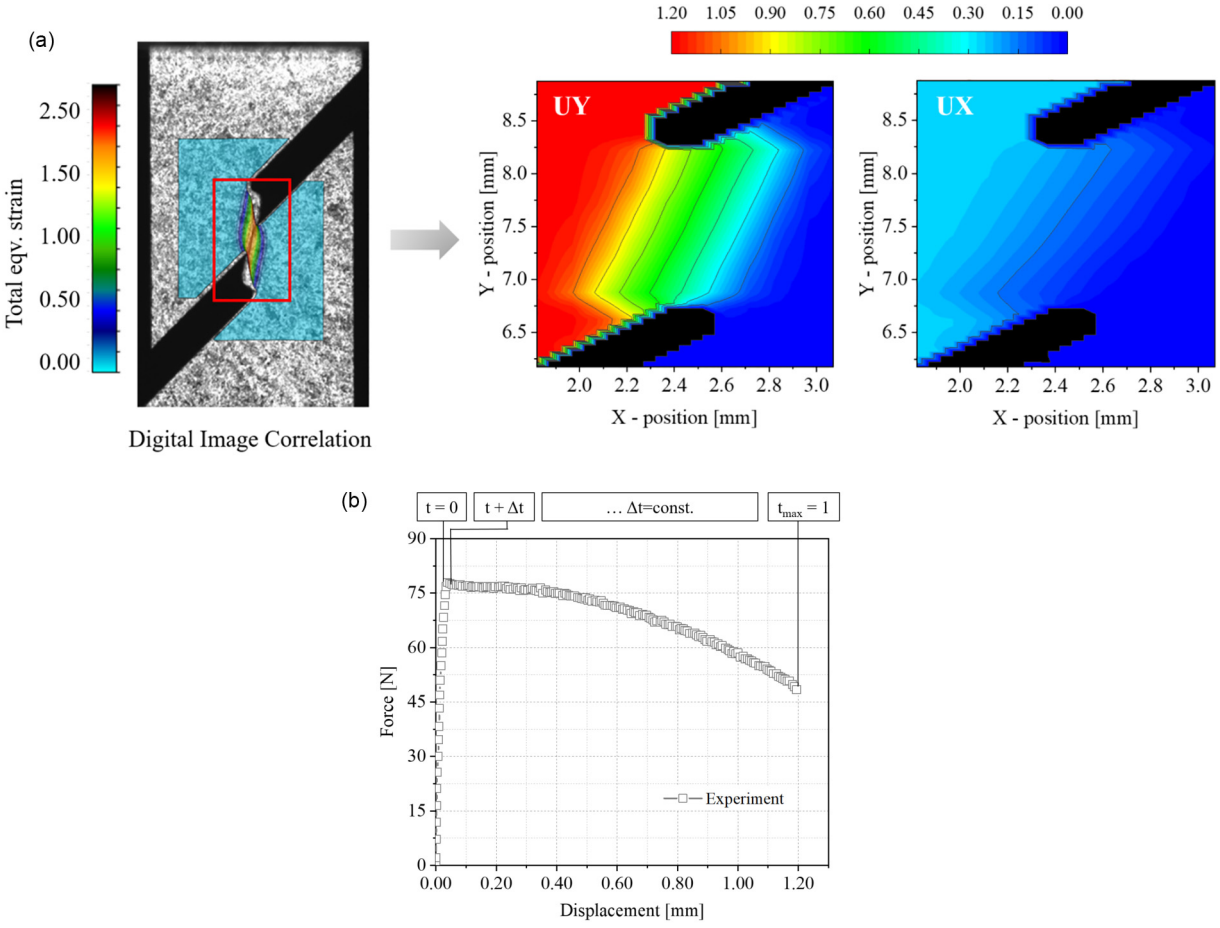


Fig. 6. Application of the trained ANN model: feeding the experimental data of the shear tension specimen into the trained ANN. (a) local displacement fields obtained from DIC analysis and (b) global force-displacement curve.

functions and are widely used in traditional ANN models. They are given by Eq. (5) and Eq. (6), respectively:

$$f(x) = \frac{1}{1+e^{-x}}, \tag{5}$$

$$f(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}. \tag{6}$$

As it is clear, the outputs of the Sigmoid are in the range (0,1) while the outputs of Tanh are in the range (−1,1). When the training data are scaled, there will be no problems with saturation which happens with large input values. For all layers in ANN structures in Fig. 4. Tanh is used except for the output layer, where Sigmoid activation is applied. Application of Sigmoid on the neurons of the last output layers imposes positivity of the material parameters $\{c_2, c_3, L\}$. Furthermore, even though the input data are scaled to (0,1) we find that Tanh is a better choice for the upper layers of the ANNs in Fig. 4: since the output of Tanh lies in the range (−1,1), the mean for the hidden layer is 0 or very close to it, which helps making the learning for next layer easier. Moreover, the Tanh activation function has larger derivatives compared to the Sigmoid, thereby making it more relevant for the gradient descent based optimization used in the error backpropagation training algorithm. Although ReLU activation functions are nowadays very popular because they overcome vanishing gradients issues, in this work, since the considered ANN models are small, i.e. there are not so many hidden layers, the ReLU functions are not applied on the hidden layers.

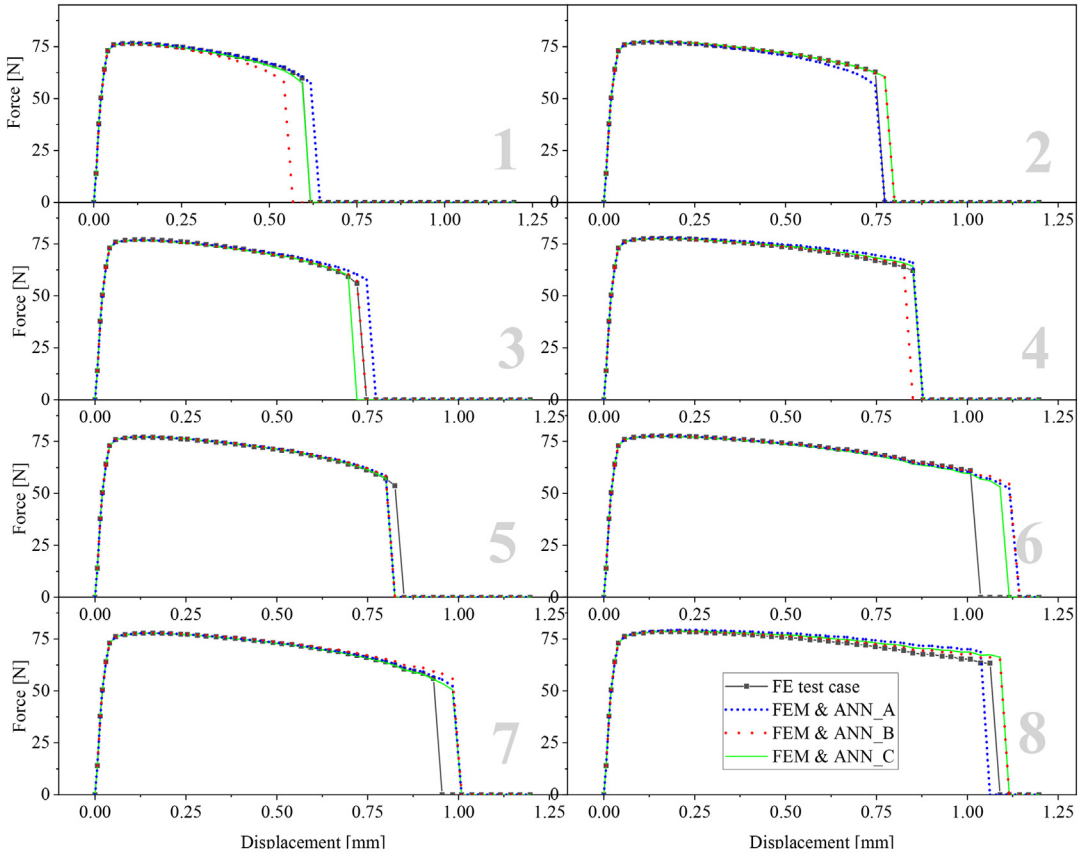
3.3.4. Network optimization

The optimization of the training parameters of the ANN models is done by minimizing the sum of squared errors (SSE) of the estimated parameters $\{c_2, c_3, L\}$ for all training examples. The SSE is defined as

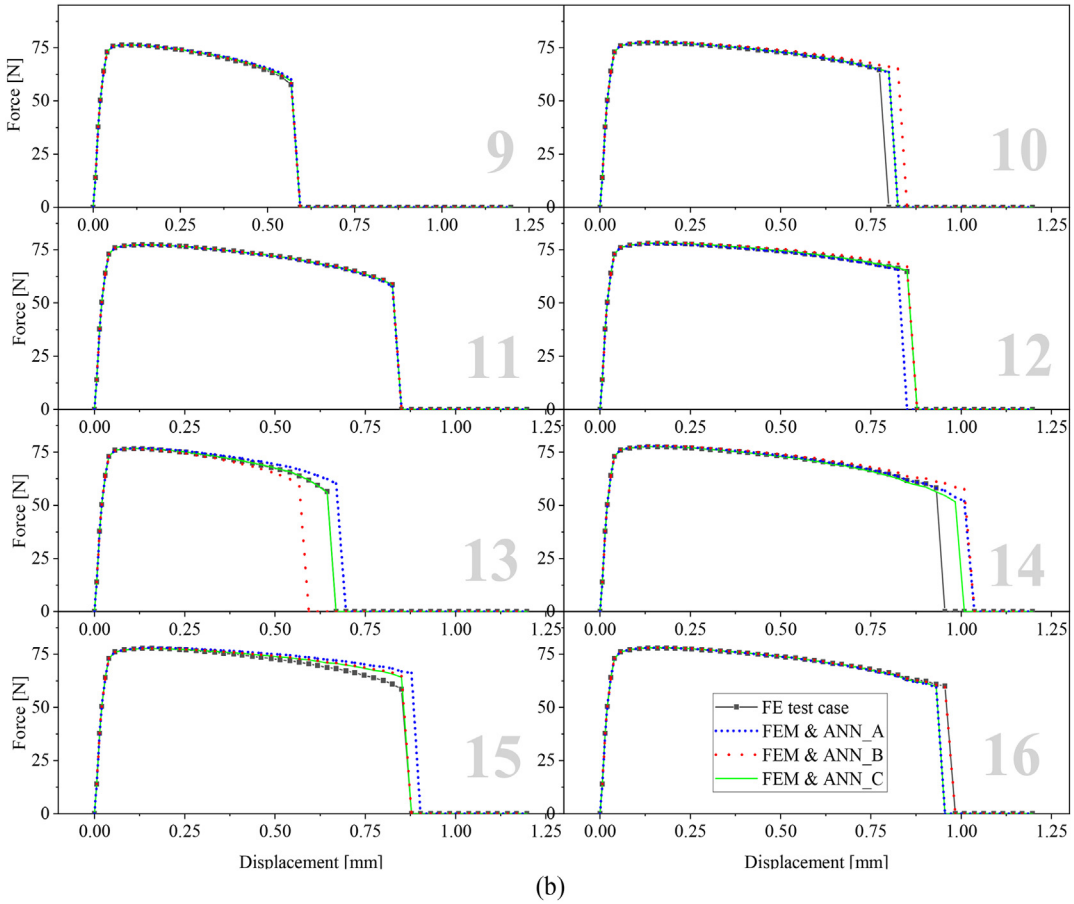
$$SSE = \sum_{i=1}^{27} \left(c'_{2i} - c_{2i}\right)^2 + \left(c'_{3i} - c_{3i}\right)^2 + \left(L'_{2i} - L_{2i}\right)^2, \tag{7}$$

where $\{c'_2, c'_3, L'\}$ are the network output and $\{c_2, c_3, L\}$ are the true target parameters. Once the networks are fed by all training examples, the accumulated error is back-propagated to update the network weights and biases. The TensorFlow® software framework is used to train and optimize the networks [1]. With TensorFlow built-in operations backpropagation based optimization can be done without the necessity to write complex program code. The Adam optimiser with a learning rate of 0.001 is used [23].

Every ANN model (A, B and C) is optimized 10 times with random initialization of its network parameters and a maximum epoch number of 4000. We found this epoch number to be sufficient for the convergence of the optimization while still preventing the model to overfit the training data due to applied regularization approaches described in Section 3.3.1. Then the average of 10 optimized model outputs is taken and considered as the final parameter estimate. With this strategy, we believe the prediction of material parameters is more accurate and robust for a highly nonlinear inverse problem as considered in this work.

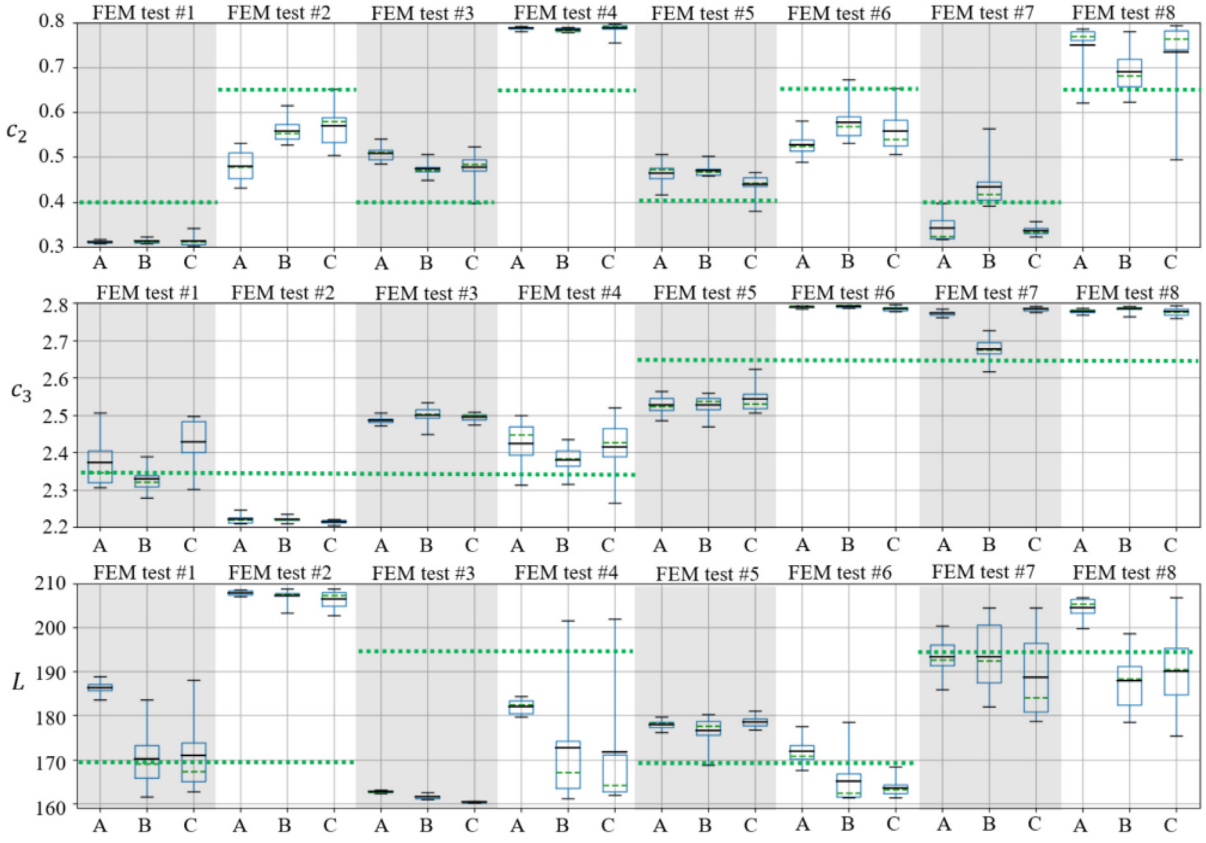


(a)

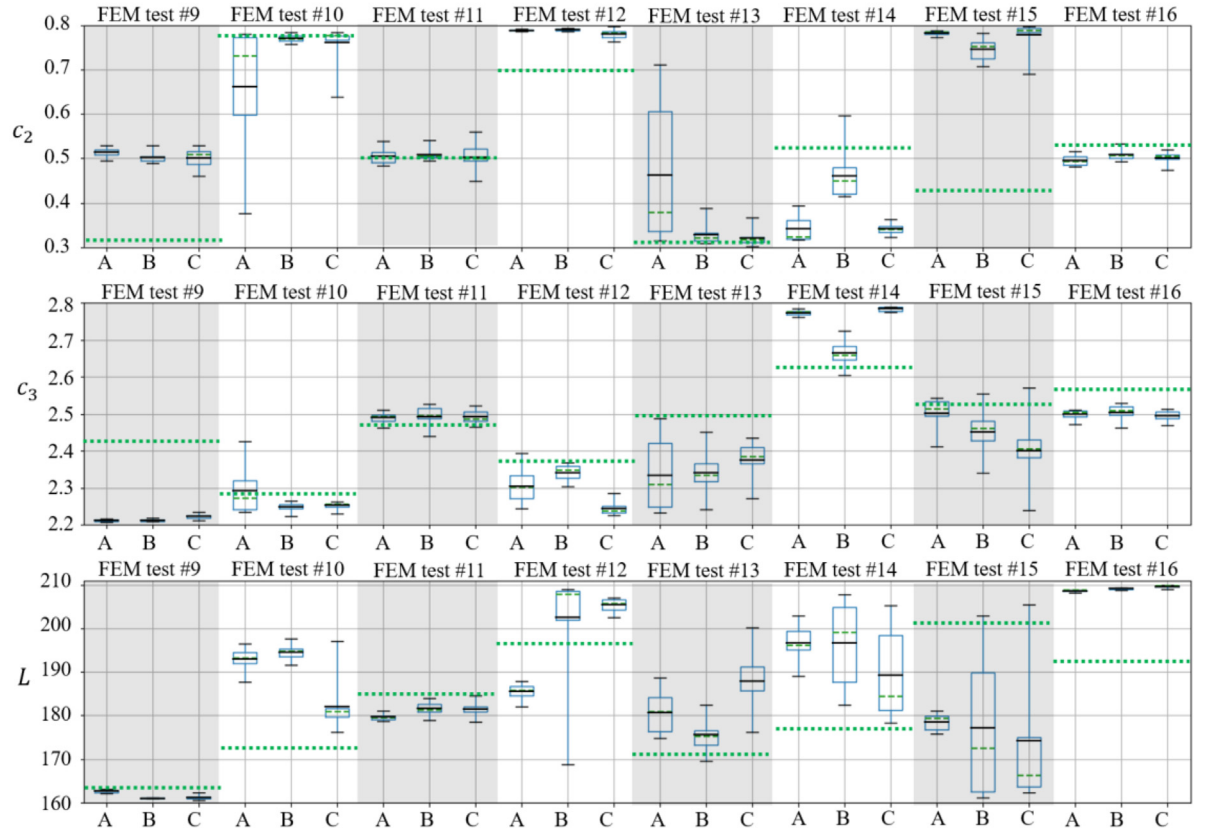


(b)

Fig. 7. Verification of the ANN models: results of the FE model predictions (FEM) for numerical test cases (FE test case) using parameter sets determined by different ANN architectures (ANN_A, ANN_B, ANN_C); the inserted numbers (1–8) on individual graph represent the number of the FE test case; (a) results for the FE models covering the middle nodes of each of the 8 cubes; (b) results for the FE models designed according to the Latin hypercube sampling.


FEM test #1
FEM test #2
FEM test #3
FEM test #4
FEM test #5
FEM test #6
FEM test #7
FEM test #8
c_2
c_3
L
A
B
C

(a)


FEM test #9
FEM test #10
FEM test #11
FEM test #12
FEM test #13
FEM test #14
FEM test #15
FEM test #16
c_2
c_3
L
A
B
C

(b)

Fig. 8. Comparative statistics of different ANN network outputs on verification data. (a) FE models covering the middle nodes of each of the 8 cubes; (b) FE models designed according to the Latin hypercube sampling. The box plots show the statistics obtained by 10 independent optimization outcomes of the ANN and the green dotted lines show the true value, used in the FE model to generate the test data, for comparison.

3.4. Feeding the experimental data to the trained ANN

The results of the tensile test conducted on the shear specimen are provided to the trained ANN in the same format as the training database. Displacement field images in X and Y directions are generated from the DIC-measured displacement history (Fig. 6a). The force values measured during tensile testing are extracted over specified time increments Δt (Fig. 6b). For every single force value, the corresponding displacement image is provided. The experimental data extracted in this way is fed to the trained ANN model to find the set of parameters $\{c_2, c_3, L\}$ that models best the behaviour of a real shear tension specimen.

When postprocessing the experimental data, different data arrangements have been generated to investigate the potential effects of the ANN predictions. The influence of spatial discretization has been assessed by extracting the displacement images in different resolutions, Δp, defined as a distance between two nodes of the FE mesh for which the data has been provided. Fine and 2× coarser resolution have been considered, $\Delta p = 0.0625$ mm and $\Delta p = 0.125$ mm, respectively. The impact of the time discretization, i.e. the time step increment Δt, has been examined as well. Finer and coarser time discretizations have been investigated, i.e. $\Delta t = 0.02$ and $\Delta t = 0.05$.

Delivering the displacement images using two different resolutions Δp has not shown any effect on the training quality. Therefore, the coarser resolution has been accepted (0.125 mm) for faster processing. On the other hand, the time discretization, i.e. the number of time steps for which the force-displacement curve data is extracted and displacement images are attached, has turned out to have some impact on the training quality. Thus, more time steps, i.e. a smaller time step increment Δt (0.02) has been adopted.

Table 4
Identified material parameters after feeding the experimental data of the shear tension specimen into ANN.

ANN architecture	Ductile damage model parameters		
	Fracture locus		Regularization
	c_2 [−]	c_3 [−]	L [MPa]
ANN_A	0.4967915	2.7909071	164.575690
ANN_B	0.4620881	2.7928969	209.217671
ANN_C	0.3201537	2.6564574	208.712200
Experimental-numerical approach from [8]	0.5570000	2.4230000	167.125000

4. Results and discussion

Before application on the real shear experiment, the quality of all three ANN architectures has been examined on four generated FE test cases unseen in the training stage. Numerical results (displacement field images and FD profiles) of those FE test cases are fed into the trained ANN models to estimate the material parameters used originally in the simulation (listed in Table 3a and Table 3b). The predicted parameters delivered by different ANN models are fed back to the FE simulation. The final model verification is based on the correct reproduction of the displacement field and FD history in the simulation with the estimated material parameters rather than merely judging the relative values of the estimated material parameters. This is because, due to the non-uniqueness of this highly nonlinear inverse problem, different parameter combinations may produce the same FD profile. Therefore, the prediction quality of the studied ANN model architectures for the FE test cases is shown in Fig. 7 by comparing the resulting FD profiles. Fig. 7a shows the predictions of the FE test cases covering the middle nodes of each of the 8 cubes. We observe that all ANN models studied (A, B and C) estimate sets of material parameters that reproduce the behaviour of most of the test cases with satisfactory accuracy. Obviously, these three ANN architectures perform very similarly. Model B is slightly off for test case 1 though. For test case 6, all three architectures have similar deviation in displacement to fracture. Fig. 7b shows the predictions of the FE test cases designed according to Latin hypercube sampling. The performance of all different ANN architectures studies is similar again for all FE test cases, with the exemption of the FE test case 13 and 14, where the ANN model B results in a somewhat larger deviation compared to ANN architectures A and C. In the light of these results it is difficult to assess the preferred model architecture. Considering the complex material model exhibiting features such as material softening, all three models provide satisfactory results.

For better evaluation of the network outputs, Fig. 8 shows comparative statistics in box plots for the outcomes of 10 independent optimized ANN models on the verification dataset. Box plots visualize the distribution of numerical data and skewness through displaying the data quartiles, averages, medians and ranges. In Fig. 8, the whiskers, i.e. upper and lower vertical lines with the horizontal cap include the total range of the

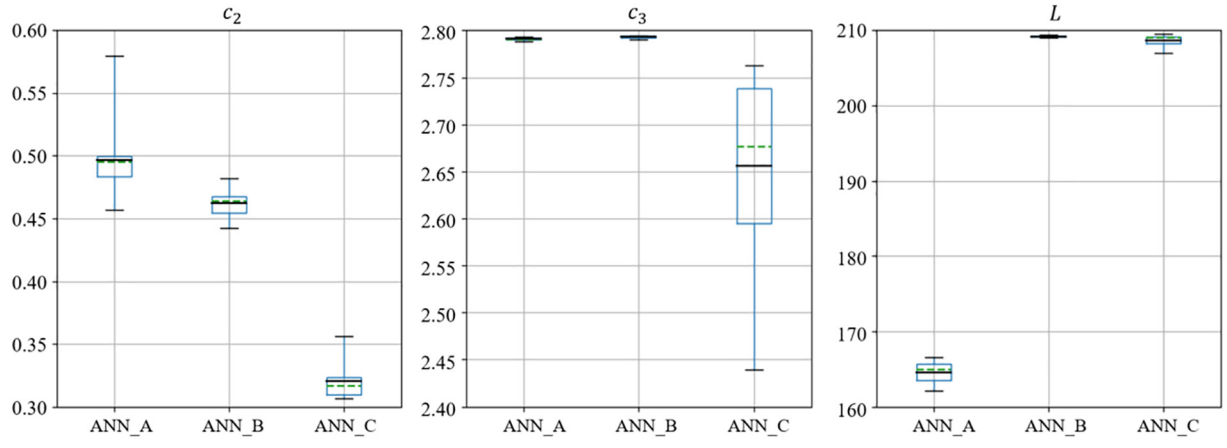


Fig. 9. Comparative statistics of different ANN network outputs on the experimental data.

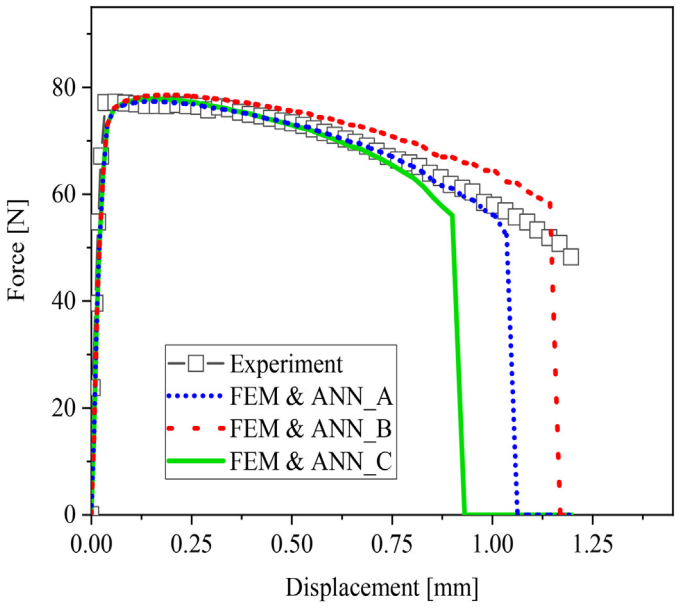


Fig. 10. Validation of the ANN model: FE model predictions of the experimental force displacement curves using material parameters determined by different ANN architectures (ANN_A, ANN_B, ANN_C).

estimates, marking the minimum and maximum of the data. The lower and upper sides of the box correspond to the first and third quartiles, respectively; the dashed line in the box corresponds to the median of the data, whereas the average of the estimates is represented as a solid line. We observe that the overall ranges of the outcomes of all models overlap most of the time. Comparing the median of each box for a given validation model of three ANN models, all architectures give results very close to each other, which reflects the fact that the models are very similar in terms of prediction quality. From the interquartile range, all models estimate the parameter c_3 in a somewhat more robust way than c_2. Furthermore, for c_3 the median and average of outcomes coincide or are very close, which reflects the Gaussian distribution of the estimates. The true values, which were input for the FE generated test data are shown as green dotted lines.

Finally, comparing the true values of the material parameters of the verification models reported in Table 3a and 3b, we observe that, although the interquartile range of the box plots rarely overlaps with the true values, the predicted parameters still produce the same numerical response (as shown in Fig. 7). Judging by the sensitivity analysis, small variations of the model parameters do not substantially change the FD curve and the displacement field. This inverse problem of estimating the material parameters is highly nonlinear, steep and complex, see Eq. (2). Therefore, it is possible to obtain different material parameters resulting in a comparable FD history and displacement field. Therefore, it is difficult to select the best model based on these verification models and exploratory data analysis. The results suggest that, in the context of the present work, all the models perform very similarly on the verification models with very small marginal differences.

However, the final validation of the ANN models has to be done on a real-world experiment. To this end, the data extracted from an experimental force-displacement curve and experimental displacement field images are fed into the trained ANN models. The predicted material parameter sets, averaged over the outcome of 10 independent optimized models, are shown in Table 4. Interestingly, we observe that models A and B predict the same parameters c_2 and c_3, unlike the prediction of parameter L which shows a considerable difference. For comparison, Table 4 shows also the parameters formerly found by a classical experimental-numerical approach [8]. Considering the complete set of parameters (c_2, c_3 and L), it becomes apparent that only the ANN model A provides comparable set of parameters. The box plot of the outcomes is shown in Fig. 9. We observe that model B has a lower scatter in predicting the parameters. Again, it is difficult to select the model which performs best.

The validation of the model prediction against experimental data is shown in Fig. 10. The parameter set provided by the ANN model A successfully predicts the force-displacement response recorded experimentally, yet with the fracture point predicted somewhat earlier than measured. On the other hand, the parameter set of the ANN model B predicts a harder response than the experimental one. The last ANN

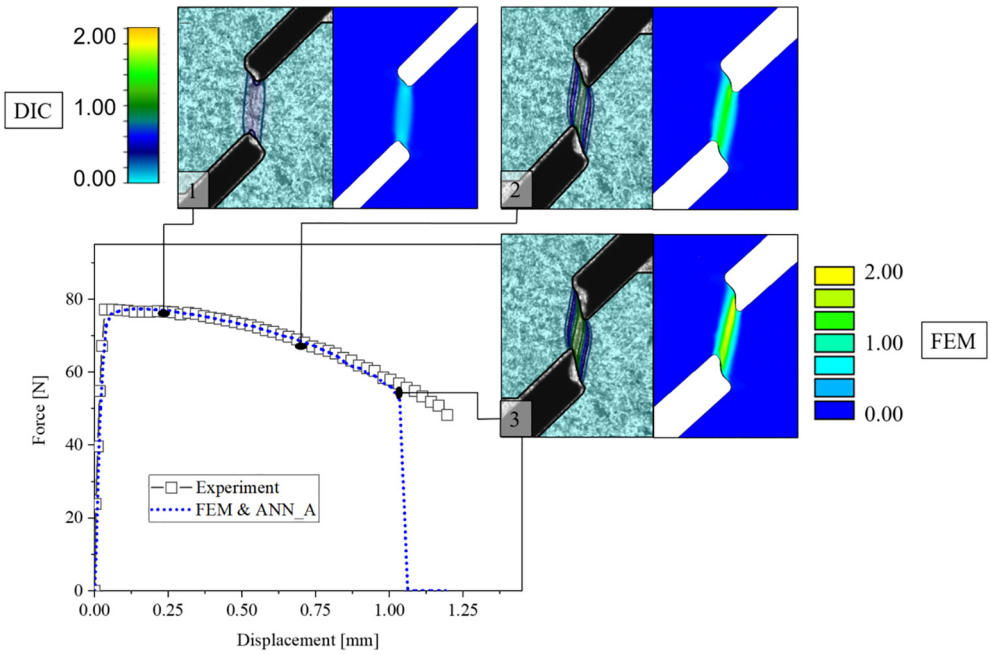


Fig. 11. Comparison of local equivalent total strain distributions documented experimentally using Digital Image Correlation (DIC) and calculated numerically (FEM) at three stages (1, 2 and 3). The FE simulation (FEM) shows the results using the set of parameters predicted by ANN model architecture A.

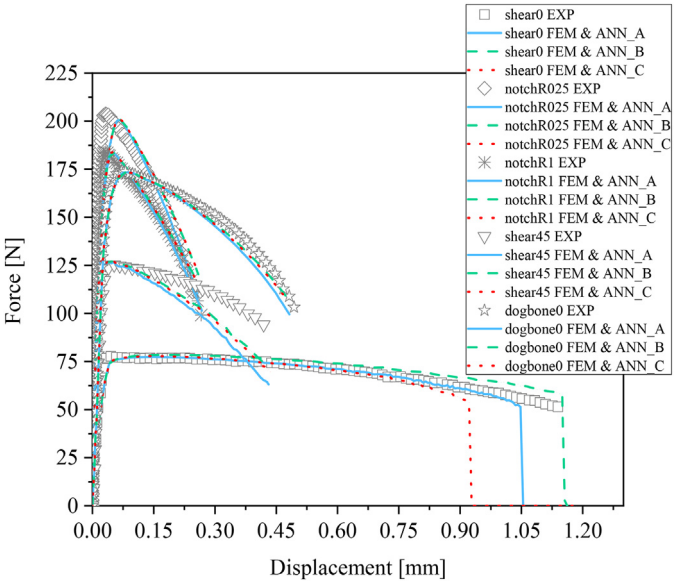


Fig. 12. Comparison of numerically predicted force-displacement curves using the set of parameters given by ANN model A for multiple geometries covering various stress states. The experimental data is taken from [8].

architecture studied, C, leads to the most conservative prediction of fracture. These results reflect the high nonlinearity of the inverse problem and show that a small change in one parameter may result in a significantly different force-displacement response. Comparing the evolution of local mechanical fields in the experiment and simulation Fig. 11, one can conclude that the progress of the strain upon loading is also well captured. This is expected because the information about the complete evolution of the local strain distribution has been accessible to the ANN in the training stage in the form of the full surface displacement field history that is fed into the ANN.

For a more general assessment of the parameter sets predicted by the ANN architectures studied, the parameters are validated for various samples not used in the training stage. The specimens specifically designed to cover various stress states in [8] are used for validation. Specimen geometries include notched samples ($R = 1.00$ mm, $R = 0.25$ mm), dogbone specimen (gauge width x length $= 2.00 \times 5.00$ mm) and shear specimen ($\alpha = 45°$) (shear specimen with $\alpha = 0°$ was used to generate the training database). Fig. 12 shows the force-displacement histories calculated for these specimens by the set of material parameters obtained from ANN architectures A, B and C. In Fig. 12, the specimens are referred to according to the list of abbreviations given in Table 5. While the numerical results are mainly following the experimental data, a somewhat larger deviation for one particular geometry (shear45) is attributed to excessive stress triaxiality influence stemming from overestimated c_2 value. Nevertheless, considering that this prediction quality is achieved using only one single specimen for the complete calibration process, the overall agreement for all specimens is satisfactory.

Table 5
List of abbreviations.

Abbreviation	Explanation
notchR025	Notched tension specimen with radius of the notch R = 0.25 mm
notchR1	Notched tension specimen with radius of the notch R = 1.00 mm
shear0	Shear tension specimen with angle α=0° (see [8])
shear45	Shear tension specimen with angle α=45° (see [8])
dogbone	Conventional dogbone tension specimen
EXP	Experimental result
SIM	Simulation result

5. Summary and conclusion

ANN assisted fracture locus calibration has been studied in this work for the first time as a promising alternative to the experimental-numerical calibration strategy followed in e.g. [8]. The article demonstrates the procedure of including the evolution of both local and global features, i.e. displacement images and force-displacement curve of the material observed in the experiment, into the training of the neural network. Application of the so-trained neural network provides accurate predictions of ductile fracture on the local and global levels, as the reported results suggest. The new methodology allows extracting the model parameters from a single sample geometry as opposed to the many sample geometries typically required to establish a ductile fracture locus. The original contributions of the presented work are twofold, stemming from the ductile fracture modelling and machine learning aspects:

- The computational framework developed supports the determination of the ductile fracture locus in such a way that the knowledge of a relevant loading path in the course of deformation is no longer mandatory for model calibration, i.e. the calibration does not rely on the strategy of averaging the stress state variables along the loading path any longer. Since it was found that the fracture locus for proportional loadings differs substantially from the one for nonproportional ones [12,14], bypassing the averaging of the fracture paths appears as a great improvement, particularly in view of the application to prestrained materials prone to strain localisation, where proportional loadings are impossible to achieve in the experiment. Furthermore, the ambiguity of the choice of a critical element to be selected in the numerical model for the extraction of the respective loading path is thereby also circumvented.
- The verification results of differently structured artificial neural networks suggest that, no matter which of these three ANN structures is chosen, it is possible to estimate realistic material parameters. Nevertheless, the preference is given to the physics informed partitioning of the ANN architecture over brute force application of ANNs. It is suggested to include prior physical knowledge regarding the material model concerned, particularly when large training data sets are not available and experimental data is limited, as it is usually the case in material modelling. This work has shown that despite limited data and thus the small size of the ANN, it is possible to calibrate the material parameters and achieve realistic results in structural computations.

Our observations suggest that a single sample geometry is sufficient to extract the constants determining the material's hardening or rather softening law. It is essential that the sample experiences a variety of different stress states in different positions upon plastic deformation. Key to success is found to be in making use of this valuable information in the model calibration. Further research should provide a more general proof involving different sample geometries and different materials. This work is a first step towards a fully automatic material parameter determination from simple tensile tests, using the evolution of local and global features as an input.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

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Appendix A

Table A.1 shows the root mean squared error (RMSE) of the model architecture C trained with different spatial kernel sizes of (3*3), (5*5), (7*7) while keeping the other hyperparameters fixed as reported. The RMSE is reported individually for each material parameter and the training and test data. A total of 10 optimization runs are performed and the average training and test error is calculated. Increasing the kernel size does improve a bit the accuracy on the training data (mainly for L) but it does not only affect the accuracy on the test data but also it deteriorates it. For small kernel size the parameter C2 has lower error on the unseen test data. Overall, we conclude putting more complexity (e.g. increasing the number of training parameters) in the architecture does not improve the model performance quality but increases the computational cost unnecessarily. The total run time (10 optimizations) on an Intel i5-6200U CPU (2.3 GHz) is also reported. We note that the effect of kernel size on the architectures A and B is similar and it is not reported here

Table A.1
Effect of the kernel size.

Kernel size = (3×3)	Training data	Middle point test data	Hypercube test data
C_2	0.027 ± 0.010	0.095 ± 0.008	0.157 ± 0.008
C_3	0.034 ± 0.005	0.125 ± 0.006	0.129 ± 0.013
L	3.424 ± 0.781	21.436 ± 1.402	16.017 ± 2.620
Kernel size = (5×5)	Training data	Middle point test data	Hypercube test data
C_2	0.027 ± 0.005	0.120 ± 0.006	0.163 ± 0.004
C_3	0.031 ± 0.006	0.114 ± 0.009	0.113 ± 0.016
L	2.492 ± 0.355	21.391 ± 1.252	16.954 ± 1.145
Kernel size = (7×7)	Training data	Middle point test data	Hypercube test data
C_2	0.021 ± 0.001	0.114 ± 0.009	0.169 ± 0.014
C_3	0.027 ± 0.005	0.113 ± 0.006	0.134 ± 0.012
L	2.244 ± 0.318	20.703 ± 1.407	16.284 ± 0.573
Kernel size	Run time (hours)		
(3×3)	38.46		
(5×5)	40.61		
(7×7)	42.54		

Table A.2 shows the training and verification accuracy of the architecture C with and without Gaussian noise considering the hyperparameters described in the manuscript. We observe that adding noise deteriorates the training accuracy slightly and on the other hand helps to generalize better on the test data, especially on the Hypercube set.

Table A.2
Effect of the Gaussian noise layer.

With Gaussian noise (0.05)	Training data	Middle point test data	Hypercube test data
C_2	0.027 ± 0.010	0.095 ± 0.008	0.157 ± 0.008
C_3	0.034 ± 0.005	0.125 ± 0.006	0.129 ± 0.013
L	3.424 ± 0.781	21.436 ± 1.402	16.017 ± 2.620
Without Gaussian noise	Training data	Middle point test data	Hypercube test data
C_2	0.021 ± 0.001	0.101 ± 0.019	0.171 ± 0.010
C_3	0.028 ± 0.004	0.109 ± 0.008	0.146 ± 0.009
L	2.710 ± 0.648	21.027 ± 0.315	16.575 ± 1.708

The optimization is more stable in the presence of a noise. Fig. A.1 shows an optimization history with and without noise considering the epoch number of 4000. Evidently, in the presence of the noise, one can stop the optimization even earlier (like at epoch of 2000) without losing accuracy. However, due to instability in the optimization without noise, we might get unlucky to stop in one of the points in which the model error has a jump. Therefore, including Gaussian noise not only improves the model generalization but also increases stability.

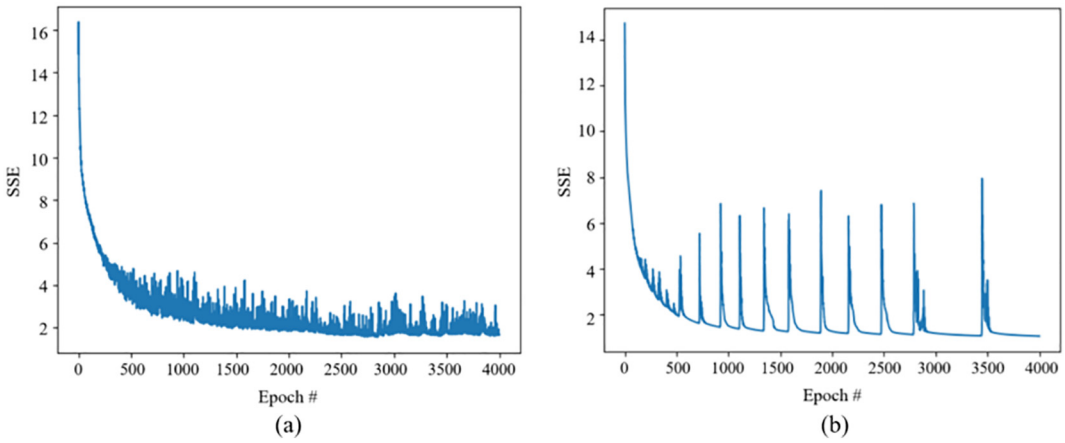


Fig. A.1. Optimization history with (a) and without (b) Gaussian noise considering the epoch number of 4000.

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